

nyquist theorem (noiseless)

$$C = 2 \times B \times \log_2 L$$

here, $L = 2^n$

$$C = 2B \log_2 2^n$$

$C \rightarrow$ Capacity

$B \rightarrow$ Bandwidth

$L \rightarrow$ no of level

Consider a noiseless channel with a Bandwidth of 3000Hz transmitting a signals with two signal level. find the Capacity of the channel? [Multiple ministry AD-2017]

$$C = 2 \times B \times \log_2 L$$

$$B = 3000$$

$$= 2 \times 3000 \times \log_2 2$$

$$L = 2$$

$$= 6000 \text{ bps}$$

$$C = ?$$

We need to send 265 kbps over a noiseless channel with a Bandwidth of 20KHz. How many signals level do we need?

$$C = 2 \times B \times \log_2 L$$

$$C = 265 \text{ kbps}$$

$$C = 265000 \text{ bps}$$

$$B = 20 \text{ kHz}$$

$$= 20000 \text{ Hz}$$

$$\Rightarrow 265000 = 2 \times 20000 \times \log_2 L$$

$$\Rightarrow 265 = 90 \times \log_2 L$$

$$\Rightarrow \log_2 L = \frac{265}{90}$$

$$\Rightarrow \log_2 L = 6.625$$

$$\therefore L = 2^{6.625}$$

$\therefore L = 98.7$ (but 98.7 is not possible because the level always 2^n)

So: $L = 64$ then the Capacity $C = 2 \times B \times \log_2 L = 2 \times 20 \times \log_2 64 = 290 \text{ kbps}$

$L = 128$ then the Capacity $C = 2 \times B \times \log_2 L = 2 \times 20 \times \log_2 128 = 280 \text{ kbps}$

$\therefore$ So we can't send 265 kbps over a noiseless

Channels, we have two options if we keep level

64 then send 290 kbps or we keep the level

128 then send 280 kbps.

Shannon theorem! (noisy channel)

$$C = B \times \log_2 (1 + \text{SNR})$$

$$C = \text{Capacity (Bps)}$$

- ① SNR এর মান প্রকৃষ্ট তা দিতে আসতে পারে।
- ② SNR এর মান dB দিতে আসতে পারে।
- ③ signal এর power এর - noise এর power (দুটো থাকতে পারে, পরে SNR এর ক্ষেত্রে)

$$B = \text{Bandwidth (Hz)}$$

$$\text{SNR} = \text{Signal to noise Ratio.}$$

✓ We can calculate the theoretical highest bit rate

of a regular telephone line. A telephone line normally

has a Bandwidth of 3000 Hz. The SNR is 3162.

Find the channel capacity.

$$C = B \times \log_2 (1 + \text{SNR})$$

$$B = 3000 \text{ Hz}$$

$$\text{SNR} = 3162 \text{ (it has no unit because it is ratio)}$$

$$C = 3000 \times \log_2 (1 + 3162)$$

$$C = 3000 \times \log_2 3163$$

$$\therefore C = 39.86 \text{ Kbps.}$$

$$C = 3000 \times 11.63$$

$$C = 34860 \text{ bps}$$

✓ The signal to noise ratio is given in decibel and that is $\text{SNR}_{\text{dB}} = 36$ and the channel Bandwidth is 2 MHz. Find the capacity of the channel.

$$\text{SNR}_{\text{dB}} = 10 \times \log_{10} \text{SNR}$$

$$\text{SNR}_{\text{dB}} = 36$$

$$B = 2 \text{ MHz}$$

$$\Rightarrow 36 = 10 \times \log_{10} \text{SNR}$$

$$\Rightarrow \frac{36}{10} = \log_{10} \text{SNR}$$

So,

$$\Rightarrow \log_{10} \text{SNR} = 3.6$$

$$\Rightarrow \text{SNR} = 10^{3.6}$$

$$\therefore \text{SNR} = 3981$$

$$C = B \times \log_2 (1 + \text{SNR})$$

$$C = 2 \times 10^6 \times \log_2 (1 + 3981)$$

$$C = 2 \times 10^6 \times \log_2 3982$$

$$C = 29 \times 10^6 \text{ bps}$$

$$C = 29 \text{ Mbps}$$

Given a channel bandwidth 3KHz and given to noise Ratio SNR is 90 dB find out Shannon capacity of this Channel and if we want to increase the bandwidth double then how much the SNR? [DPDE-2025]

$$\text{SNR}_{\text{dB}} = 10 \times \log_{10} \text{SNR}$$

$$\Rightarrow 90 = 10 \times \log_{10} \text{SNR}$$

$$\Rightarrow \log_{10} \text{SNR} = 9$$

$$\therefore \text{SNR} = 10^9 \Rightarrow 1000000000$$

$$B = 3 \text{ KHz} = 3000 \text{ Hz}$$

$$\text{SNR}_{\text{dB}} = 90 \text{ dB}$$

if Bandwidth double ($B = 6000 \text{ Hz}$)

find new SNR in dB to maintain the same capacity

$$C = B \times \log_2 (1 + \text{SNR})$$

$$C = 3000 \times \log_2 (1 + 10000)$$

$$C = 3000 \times \log_2 (10001)$$

$$C = 3000 \times \frac{\log_{10}(10001)}{\log_{10}(2)}$$

$$C = 3000 \times \frac{9.000093}{0.3010}$$

$$C = 3000 \times 13.29$$

$$C = 39870 \text{ bps} \approx 39.87 \text{ kbps}$$

⇒ Double the bandwidth, keep capacity same

$$B' = 6000 \text{ Hz}, \text{SNR} = x$$

$$C = 6000 \times \log_2 (1 + x)$$

$$\Rightarrow \log_2 (1 + x) = \frac{39870}{6000}$$

$$\Rightarrow \log_2 (1 + x) = 6.645$$

$$\Rightarrow 1 + x = 2^{6.645}$$

$$\therefore x = 99.03$$

$$\therefore \text{SNR}_{\text{dB}} = 10 \times \log_{10} \text{SNR}$$

$$= 10 \times \log_{10} 99.03$$

$$= 19.96 \text{ dB}$$

$$\approx 20 \text{ dB}$$

A communication channel has bandwidth 1 MHz and operation in AWGN with SNR = 20 dB calculate the maximum data rate using Shannon capacity theorem [uasa-2025]

$$\text{SNR}_{\text{dB}} = 10 \cdot \log_{10} \text{SNR}$$

$$\Rightarrow \log_{10} \text{SNR} = \frac{\text{SNR}_{\text{dB}}}{10}$$

$$\Rightarrow \text{SNR} = 10^{\frac{\text{SNR}_{\text{dB}}}{10}}$$

$$\Rightarrow \text{SNR} = 10^{\frac{20}{10}}$$

$$\Rightarrow \text{SNR} = 10^2 \therefore \text{SNR} = 100$$

$$B = 1 \times 10^6 \text{ Hz}$$

$$\text{SNR} = 20 \text{ dB}$$

the channel capacity

$$C = B \log_2 (1 + \text{SNR})$$

$$C = 17 \times 10^6 \times \log_2 (1 + 100)$$

$$C = 10^6 \times \log_2 101$$

$$C = 10^6 \times \frac{\log_{10}(101)}{\log_{10}(2)}$$

$$C = 10^6 \times 6.657$$

$$C = 6.657 \text{ bps}$$

this is the maximum theoretical
data rate under ideal
conditions in the presence
of Additive white Gaussian
noise (AWGN)